LeSuccess Academy

Learn . Educate . Succeed | Official Course Syllabus

DSA with Python / Java

Duration: 3-6 Months | Mode: HYBRID | 100% Placement Support

Detailed Curriculum & Module Outline:

Module 1: Complexity Analysis and Recursion

Big-O in practice, recursion, call stacks and base cases - the vocabulary the whole course uses

Module 2: Arrays and Strings

Two pointers, sliding window, prefix sums and the in-place tricks these problems keep asking fo

Module 3: Linked Lists

Singly, doubly and circular lists, reversal, cycle detection and merging, written without leaki

Module 4: Stacks and Queues

Monotonic stacks, deques and the parsing and next-greater-element problems built on them

Module 5: Hashing and Hash Maps

Hash tables, collisions, and the counting and lookup problems where a map turns O(n) into O(n)

Module 6: Trees and Binary Search Trees

Traversals, height, balanced trees and the recursive patterns most tree questions reduce to

Module 7: Heaps and Priority Queues

Top-K, running median and scheduling problems, and when a heap beats a sort

Module 8: Graphs

Representations, BFS, DFS, topological sort and shortest paths, applied to grid and network pro

Module 9: Sorting and Searching

Merge sort, quick sort, binary search and the "search the answer" pattern that hides inside man

Module 10: Greedy Algorithms

Interval scheduling and selection problems, and how to tell when greedy is provably wrong

Module 11: Dynamic Programming

Memoisation, tabulation, knapsack, subsequences and grid paths, built up from recursion rather

Module 12: Mock Interviews and Contest Practice

Timed rounds on LeetCode and HackerRank with live review of how you approached, not just what y